Printed Pages: 02 **Sub Code:KOE044**

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B.TECH.
(SEM IV) THEORY EXAMINATION 2022-23
SENSORS & INSTRUMENTATION

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If you require any missing data, then choose suitably.

SECTION A

1. Attempt*all* questions in brief. 2x10 = 20

(a) Define transducers.
(b) Define stress and strain?
(c) Discuss primary and secondary transducer.
(d) Define proximity sensor.
(e) Define sensitivity and resolution of ADC.
(f) Describe thermal imaging.
(g) Write application of digital counter.
(h) Discuss clusters and arrays?
(i) Define data socket and their importance for network communication.
(j) Give two advantages Intelligent sensor.

SECTION B

2. Attempt any *three* of the following: 10x3 = 30

(a) Explain measurement of force using strain gauge.
(b) Illustrate sigma delta type analog to digital converter.
(c) Define structure in Lab View in Virtual instrumentation. Discuss Flat sequence structure in Lab View.
(d) Design 3-bit asynchronous counter.
(e) Explain in detail.
    (i) RTD
    (ii) Thermistor

SECTION C

3. Attempt any *one* part of the following: 10x1 = 10

(a) A linear resistance potentiometer is 5 cm long and uniformly wound with a wire having a resistance of 10 KΩ. Under normal conditions, the slider is at the center of potentiometer. What still the linear displacement be when the resistance of potentiometer as measured by bridge circuit at i) 3.8 kΩ ii)8.3 kΩ
(b) Explain working of Optical Encoder and write one application of optical encoder.

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4. Attempt any *one* part of the following: 10 x1 = 10

(a) Discuss the measurement of pressure using LVDT based diaphragm.

(b) A platinum resistance thermometer has a resistance of 100 Ω at 0°C. What is the resistance when the temperature is 150°C? When the temperature has a resistance of 250Ω, what is the value of temperature? Platinum has a resistance temperature coefficient of 0.0039/°C.

5. Attempt any *one* part of the following: 10x1 = 10

(a) Write an example of case and sequence structure in graphical programming.

(b) Discuss DAQ in details with its type.

6. Attempt any *one* part of the following: 10x1 = 10

(a) Discuss Successive approximation types (SAR) ADC with one example.

(b) Explain working of binary weighted resistor type digital to analog converter.

7. Attempt any *one* part of the following: 10x1 = 10

(a) Illustrate LabVIEW. Discuss different windows associated with it. Write its advantages.

(b) Discuss application of smart sensors as automatic engine control.

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